

RESEARCH INTERESTS

Computational Geomechanics | Poromechanics | Reservoir Simulation | Scientific Machine Learning

EDUCATION

Ph.D. in Earth Science - Cornell University	2024 - Present
Minors: Computer Science and Scientific Computing	
B.S. in Mining Engineering - Northeastern University (China)	2024

RESEARCH EXPERIENCE

Ph.D. Student - Department of Earth and Atmospheric Sciences, Cornell University	Aug. 2024 - Present
Advisor: Chloé Arson Ithaca, New York	
<i>Computational geomechanics and poromechanics for coupled multiphysics problems in subsurface and reservoir systems, supported by physics-based and machine-learning surrogate modeling.</i>	
- Develop coupled poromechanics and computational geomechanics models of subsurface and reservoir systems in the MOOSE finite element framework, including hydraulic fracturing and geothermal injection-production. - Build physics-informed neural networks, implicit neural representations, and operator-learning surrogates for geometry-aware and physics-aware prediction at a fraction of full simulation cost. - Study multiscale transport in porous media through homogenization theory and investigate AI-driven discovery of interpretable upscaled permeability laws. - Lead cross-institution work on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.	
Undergraduate Researcher - Northeastern University (China)	Sep. 2020 - Jun. 2024
B.S. in Mining Engineering Shenyang, China	
- Conducted research in mining geomechanics spanning hydraulic fracturing, triaxial rock testing, tunnel boring machines, and AI-driven microseismic processing. - Developed a unified deep-learning model for microseismic signal classification and arrival-time picking; received the Outstanding Thesis Award.	

RESEARCH SUPPORT AND COMPUTATIONAL ALLOCATIONS

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs	2026
National Science Foundation (NSF), ACCESS program - Awarded. Student lead: Y. Liu; Advisor: C. Arson. Total allocation and Arson group share: 1.5M CPU hours.	
AI-aided Computational Geomechanics	2026
Empire AI Alpha+Beta - Awarded. PIs: Y.C. Han, Y. Liu, A. Tristani, C. Arson. Total allocation and Arson group share: 5.5K GPU hours.	
A Physics-Guided Subsurface Earth-System Model for Uncertainty-Aware Prediction of Geothermal Reservoir Evolution	2026
NASA ROSES-25 F.5 FINESST - Under review. Future Investigator: Y. Liu; PI / faculty mentor: C. Arson.	

PUBLICATIONS

- **Yulong Liu**, Chloé Arson. *A Physics-Informed Neural Network for Modeling Pressurized Cavities of Arbitrary Smooth Shape Embedded in Heterogeneous Rock*. **Accepted in Rock Mechanics and Rock Engineering**, 2026. Research Square preprint: doi:10.21203/rs.3.rs-8492281/v1.
- **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection-Production Systems: A Cornell Case Study*. ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, 2026.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling*. Tunnelling and Underground Space Technology, 172, 107501, 2026.
- **Yulong Liu**, Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media*. ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Yulong Liu**, Zhenghan Song, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code*. arXiv preprint; under review at NeurIPS 2026.

- Zhenghan Song, Yunyi Li, **Yulong Liu**. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking*. Under review at ACL ARR 2026.

THESIS

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning*. Undergraduate thesis, Northeastern University, 2024.

CONFERENCES AND PRESENTATIONS

Oral presentation , 60th US Rock Mechanics/Geomechanics Symposium (ARMA), Tucson, AZ	Jun. 2026
Invited speaker and panelist , 20th Phase Field Methods Workshop, Northwestern University	May 2026
Poster presentation , Cornell CEE Graduate Research Symposium, Ithaca, NY	Apr. 2026
Invited seminar , Physics-Informed AI Optimization of Heterogeneous Rock Systems, Southwest Petroleum University	Dec. 2025
Poster presentation , 59th US Rock Mechanics/Geomechanics Symposium (ARMA), Santa Fe, NM	Jun. 2025

FELLOWSHIPS AND AWARDS

Estwing Hammer Award - Outstanding EAS Graduate Student of 2024, Cornell University	2025
Cornell University Travel Grant	2025, 2026
Outstanding Thesis, Northeastern University	2024
National Scholarship, Ministry of Education, China	2024
First Class Scholarship, Northeastern University	2023
Autumn Scholarship; Golden Seed Scholarship, Northeastern University	2022

ACADEMIC SERVICE

- Reviewer, Rock Mechanics and Rock Engineering
- Reviewer, Underground Space
- Reviewer, International Conference on Artificial Neural Networks
- Reviewer, Geothermal Rising Conference

SKILLS

Programming & Scientific Computing	Python, C++, R, MOOSE finite element framework
Modeling & Simulation	Finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD
Subsurface & Reservoir	Hydraulic fracturing, geothermal injection-production, multiscale transport in porous media
Scientific Machine Learning	ML for PDEs, PINNs, operator learning, implicit neural representations, surrogate modeling
LLMs for Simulation	Simulation-software workflows and physical-intent verification of generated code
Experimental Methods	Rock mechanics testing, triaxial compression, laboratory instrumentation
Scientific Writing	LaTeX / Overleaf

PROFESSIONAL MEMBERSHIPS

- American Rock Mechanics Association (ARMA), 2025-Present
- International Society for Rock Mechanics and Rock Engineering (ISRM), 2025-Present
- Chinese Society for Rock Mechanics and Engineering (CSRME), 2024-Present
- Geothermal Rising, 2025-Present